

# Case morphology and word order in second language Turkish: Evidence from Greek learners

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## Abstract

The optional use of morphology attested in second language learners has been attributed either to a representational deficit or to a 'surface' problem with respect to the realization of inflectional affixes. In this article we contribute to this issue by providing empirical data from the early interlanguage of Greek learners of Turkish. Three experiments have been conducted, a cloze task, a sentence picture matching task and an on-line grammaticality judgement task, in order to investigate case morphology and its interaction with word order constraints. The findings of all three experiments point towards a variable use of case morphology, which is also observed in previous studies of Turkish as a second language (L2). Moreover, they show clearly that the learners face difficulties with non-canonical word orders as well as with the interaction of word order constraints and Case. On the other hand, the learners performed well on verbal inflections. On the basis of these findings, we argue that the developmental patterns in the early stages of L2 acquisition cannot be attributed to a global lack of functional categories but rather

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to more localized difficulties, which seem to be related to (a) whether the features in the L2 are grammaticalized in the first language and (b) the way these features are encoded in the morphosyntax of the first language. Moreover, we claim that processing factors and the specific properties of the morphological paradigms affect L2 development.

### Keywords

second language acquisition, Turkish, case morphology, word order, specificity

## I Introduction

One of the most robust observations in the field of first-language (L1) acquisition is that verbal and nominal inflection is omitted frequently in the early child grammar. Inconsistent use of inflection has also been observed in the grammars of second-language (L2) learners. This variability or optionality in the use of inflection in the L1 and L2 grammars has provoked an ongoing debate in the two fields regarding the status of functional categories and features associated with the missing inflectional elements.

As far as L1 acquisition is concerned, there are two contrasting views on morphological variability. According to ‘maturational’ accounts (cf. Radford, 1990), the absence of overt morphology in the child data is taken as evidence for the absence of associated functional categories, which are supposed to mature based on biologically determined factors. According to ‘continuity’ accounts, however, components of Universal Grammar (UG) are present at the early stages of acquisition and develop based on the interaction between UG and the incoming input. For ‘weak continuity’ approaches, functional projections are not fully accessible, although some of them may be available at the onset of acquisition (cf. Clahsen, 1991; Vainikka, 1993/94). Finally, ‘strong continuity’ approaches assume that all functional categories are available from the early stages of acquisition but they are underspecified (Poeppl and Wexler, 1993; Wexler, 1994).

With respect to L2 acquisition, it has been argued that variability in the use of inflection indicates that the associated functional categories are missing from the learners’ grammars (Vainikka and Young-Scholten, 1994, 1996, 2006, 2007, 2009). Other researchers, however, argue that functional categories are present in L2 grammars and the optionality in the use of inflection is due to problems associated with surface morphology (Haznedar and Schwartz, 1997; Lardiere, 1998a, 1998b; Prévost and White, 2000, among others). In contrast to the above accounts, other studies suggest that inflectional variability is associated only with those features that are uninterpretable and not grammaticalized in the L1 (cf. Hawkins and Chan, 1997; Tsimpli, 2003; Hawkins and Hattori, 2006; Tsimpli and Dimitrakopoulou, 2007; Tsimpli and Mastropavlou, 2008).

In this article we contribute to this debate by investigating the acquisition of case suffixation and its interaction with word order constraints in the grammar of Greek students learning Turkish as a foreign language. Our results replicated previous findings in that L2 learners of Turkish show optionality in the use of case suffixes. However, they experience difficulties with the interaction of word order constraints and case morphology. On the other hand, our learners display good performance on verbal suffixes.

The article is organized as follows: Section II presents an overview of L2 theories on the variability of inflection and the availability of functional categories in L2 grammars. The structure of Turkish and Greek with specific reference to case morphology and (non)specific objects is discussed in Section III. Section IV offers a summary of studies on the acquisition of case inflection in Turkish as an L2. Section V presents the methodology of the study, followed by the results on case suffixation and its interaction with word order constraints in Section VI. The implications of this study for L2 theories are discussed in the final section.

## II Theoretical perspectives on L2 acquisition

The availability and status of functional categories related to verbal and nominal inflection has been debated among researchers of L2 acquisition. The basis for this debate lies on the variability or optionality that L2 learners exhibit in the use of verbal (e.g. tense, agreement) and nominal (e.g. case, gender) morphology as well as in the use of functional elements like auxiliaries, complementizers and determiners. The debate has focused on whether functional categories are present at the onset of the L2 acquisition process or whether they develop later.

According to a number of researchers, functional categories are initially absent in the grammars of L2 learners and develop in succession during the acquisition process due to positive evidence from the morphological paradigm of the input language. For example, based on the lack of verb raising, auxiliaries and modals, complementizers and *wh*-movement as well as on the impoverished agreement paradigm in their Korean and Turkish L2 speakers of German, Vainikka and Young-Scholten (1994, 1996) argue that only lexical categories are present at the earliest stage of L2 acquisition. The subsequent acquisition of functional projections is input-driven resulting in an early underspecified functional projection (FP), which is gradually replaced by a fully specified IP/AgrP, which is in turn followed by a CP level. According to ‘organic grammar’, a more recent reformulation of Vainikka and Young-Scholten’s account, early L2 grammars include minimal trees, which contain only lexical categories that are L1-based. The minimal trees in the L2 become L2-target based on the available evidence from the input (Vainikka and Young-Scholten, 2006, 2007, 2009).

In contrast, other L2 theories propose that all functional projections are transferred from the L1 to the L2 already at the initial stage and that the L2 development is guided by UG (Full Transfer/Full Access hypothesis; Schwartz and Sprouse, 1994, 1996). According to this approach, morphological variability in the L2 grammar does not indicate an impairment in the underlying representations of L2 learners but a problem in the realization of surface morphology, that is a mapping problem between syntactic projections/features and morphological forms (cf. so-called ‘morphological deficit’ accounts; Lakshmanan, 1993/94; Gavrusseva and Lardiere, 1996; Grondin and White, 1996; Haznedar, 1997; Haznedar and Schwartz, 1997; Lardiere, 1998a, 1998b; White, 1998; Prévost and White, 2000; Haznedar, 2006). Based on the presence of the Determiner Phrase, Inflectional Phrase and Complementizer Phrase in adult and child L2 grammars, such approaches have argued that functional projections are present from the earliest stages of L2 acquisition. For example, Gavrusseva and Lardiere (1996) present evidence from child L2 English for the early emergence of CP-related material (inversion in yes/no questions and *wh*-questions)

and for nominative pronominal subjects despite the variability of inflection associated with the IP projection (see also Haznedar and Schwartz (1997) for similar evidence from child L2 English). On similar grounds, Grondin and White (1996) present evidence that Det, Infl and probably Comp are present in child L2 French based on the consistent use of determiners and pronominal possessives, overt case assignment in DPs, placement of negation, adverbs, quantifiers as well as subject and object clitics. Prévost and White (2000) also argue that interlanguage grammars do not suffer from a syntactic deficit but rather from a mapping problem between syntactic features and their morphological realization. They show that adult learners of French and German, despite their optional use of finite verb forms, manifest internal knowledge of 'finiteness', as they do not use finite forms in non-finite contexts and their inflected forms are correctly marked for agreement. In a similar vein, White (2003) reports on production data from a Turkish learner of English, which demonstrate high level of accuracy in the verbal domain and relatively lower level in the use of definite and indefinite articles. Moreover, based on the fact that the syntactic properties associated with the verbal (i.e. argument drop, verb placement) and the nominal (i.e. +/- definiteness) domains are not impaired, White argues that the optionality attested in the L2 data is due to problems with morphological realization rather than with underlying syntactic representations. Furthermore, with respect to final stages of L2 development, Lardiere (1998a, 1998b) shows that despite the deficient use of the 3rd person singular and the past tense suffixes in adult L2 English, there is evidence for Agreement and Tense at an abstract level due to the suppli-

ance of nominative subjects and correct adverb placement.

As far as the final L2 stages are concerned, theories such as the Failed Functional Features hypothesis (Hawkins and Chan, 1997; Hawkins and Hattori, 2006) and the Interpretability hypothesis (Tsimplici, 2003; Tsimplici and Dimitrakopoulou, 2007) assume a more localized representational deficit in L2 grammars and, in particular, one that affects uninterpretable features that are not grammaticalized in the L1. More specifically, Tsimplici and Dimitrakopoulou (2007) compared the performance of Greek speakers of English on acceptable gaps and unacceptable resumptive pronouns in *wh*-questions and concluded that the resumption strategy is in operation in the L2 due to L1 interference and it is affected by the interpretability of the features involved. Moreover, Hawkins and Hattori (2006) investigated the interpretation of bi-clausal *wh*-questions by Japanese speakers of English and found that the uninterpretable feature involved in English *wh*-questions, which is absent in Japanese, was not fully acquired by the learners.

The aim of the present study is to investigate case suffixation and its interaction with word order in Greek learners of Turkish. More specifically, we conducted three tasks, a cloze task, a sentence-picture matching task and an on-line grammaticality judgement task, in order to examine whether adult Greek learners of Turkish correctly supply case suffixes and show sensitivity to word order constraints.

### III Case and (non)specific objects in Turkish and Greek

Both Greek and Turkish employ morphological case in order to mark certain grammatical functions and categories. In what follows, we discuss the case systems of the two languages but we focus primarily on the cases associated with the subject and the direct object of main clauses which is the topic under investigation in this article.

## 1 Greek

Greek distinguishes three cases: nominative is the case for the subject (1), accusative prototypically marks the direct object (2) and genitive mainly marks oblique arguments (3a) and adnominal functions, such as the possessive (3b), etc. (Holton et al., 1997):

- (1) o            janiis            kimate.  
the-NOM John-NOM sleep-3SG  
'John is sleeping.'
- (2) o            filakas            xtipise            to            jani.  
the-NOM guard-NOM hit-PAST.3SG the-ACC John-ACC  
'The guard hit John.'
- (3) a. i            maria            eðose            tu            jani            ena            fili.  
the-NOM Maria-NOM give-PAST.3SG the-GEN John-GEN a-ACC kiss-ACC  
'Maria gave a kiss to John.'
- b. o            fakos            tu            filaka.  
the-NOM torch-NOM the-GEN guard-GEN  
'The guard's torch.'

Moreover, Greek employs prepositional phrases to express the indirect object (4) as well as functions such as origin/source (5) and location (6).<sup>1</sup>

- (4) o            ali            eðose            sto            xasan            to            vivlio.  
the-NOM Ali-NOM give-PAST.3SG to.the-ACC Hasan-ACC the-ACC book-ACC  
'Ali gave the book to Hasan.'
- (5) o            xasan            irðe            apo            tin            agira.  
the-NOM Hasan-NOM come-PAST.3SG from the-ACC Ankara-ACC  
'Hasan came from Ankara.'
- (6) to            vivlio            ine            sto            trapezi.  
the-NOM book-NOM is on-the-ACC table-ACC  
'The book is on the table.'

Greek is a synthetic language with a great degree of fusion in the marking of grammatical categories. Case is fused with number and there is extensive lexically determined allomorphy depending on the inflectional class the relevant noun belongs to (Ralli, 2000). Thus, for the same case, there are different formatives for the singular and the plural (andras-NOM.SG 'man', andres-NOM.PL 'men') as well as for different inflectional classes (andras-NOM.SG 'man', naftis-NOM.SG 'sailor').

## 2 Turkish

Turkish morphology distinguishes six cases, namely nominative, accusative, genitive, dative, ablative and locative. Their main functions are the following (Kornfilt, 1997; Göksel and Kerslake, 2005):

- (7) *nominative: subject*  
 Hasan-Ø uyan-dı.  
 Hasan-NOM wake up-PAST  
 'Hasan woke up.'
- (8) *accusative: object*  
 Ali Hasan-ı gör-dü.  
 Ali Hasan-ACC see-PAST  
 'Ali saw Hasan.'
- (9) *genitive: possession*  
 Hasan-ın kitab-ı  
 Hasan-GEN book-3SG  
 'Hasan's book'
- (10) *dative: indirect object, direction*  
 Ali kitab-ı Hasan-a ver-di.  
 Ali book-ACC Hasan-DAT give-PAST  
 'Ali gave the book to Hasan.'
- (11) *ablative: origin, source*  
 Hasan Ankara-dan gel-di.  
 Hasan Ankara-ABL come-PAST  
 'Hasan came from Ankara.'
- (12) *locative: location*  
 kitap masa-da.  
 book table-LOC  
 'The book is on the table.'

The morphological marking of cases is transparent in Turkish since it is an agglutinative language. Thus, the category of case is marked with distinct specialized suffixes that appear at the end of the nominal element. Case allomorphy is always phonologically determined, since the vowel of these suffixes is subject to rules of vowel-harmony, whereas the presence of an initial consonant or its voicing value depends on the featural composition of the final phoneme of the preceding root (Lewis, 2000; Göksel and Kerslake, 2005). The formatives for each case and the phonological shapes they take are given in (13):

- (13) a. nominative  
 -Ø  
 b. accusative  
 -(y)ı → -(y)i, -(y)u, -(y)ı, -(y)ü  
 c. genitive  
 -(n)ın → -(n)in, -(n)un, -(n)ın, -(n)ün  
 d. dative  
 -(y)A → -(y)a, -(y)e,  
 e. ablative  
 -DAn → -dan, -den, -tan, -ten  
 f. locative  
 -DA → -da, -de, -ta, -te

In Turkish, subjects are in nominative case, which is marked with the default null formative (7), whereas direct objects are normally marked with the accusative suffix *-(y)I* (8). However, there is a complication about the accusative marking of objects in the sense that Turkish is a Differential Object Marking (DOM) language. In DOM languages, an object may or may not be case marked depending on its semantic and/or pragmatic features (Comrie, 1979, 1989; Silverstein, 1981; Bossong, 1985; Croft, 1988; Lyons, 1999; Aissen, 2003, among others). Turkish exhibits DOM with respect to specificity (Enç, 1991; Kornfilt, 1997; Lyons, 1999; Aissen, 2003; von Heusinger and Kornfilt, 2005, among others). Thus, specific objects (definite and indefinite) are marked by the accusative marker *-(y)I*, whereas non-specific and non-referential (generic) objects carry no overt case morphology:<sup>2,3</sup>

- (14) a. Ali bir kitab-ı al-dı. (indefinite, specific)  
 Ali a book-ACC buy-PAST  
 'A book is such that Ali bought it.'  
 b. Zeynep adam-ı gör-dü. (definite, specific)  
 Zeynep man-ACC see-PAST  
 'Zeynep saw the man.'
- (15) a. Ali bir kitap al-dı. (indefinite, non-specific)  
 Ali a book buy-PAST  
 'Ali bought some book or other.'  
 b. Bilet sat-ıyor-lar. (non-referential)  
 ticket sell-PRES-PL  
 'They are selling tickets.'

It should be noted that DOM does not apply in constructions where the object receives lexical dative or ablative case. In these situations, the object always appears in the relevant case regardless of whether it is specific or non-specific:

- (16) Ahmet at-a bin-er.  
 Ahmet horse-DAT ride-HABIT  
 'Ahmet rides the horse.' or 'Ahmet is a horse-rider.'

Finally, case and specificity also affect word order in Turkish.<sup>4</sup> Turkish is a head final language, hence a head always follows its complements. This means that objects precede verbs, NP-complements precede their nominal heads, adpositions always appear as postpositions, etc. Given this typology, the basic word order for Turkish is SOV (see examples 8, 10, 14–17) (Erguvanlı, 1984). However, the order between S and O is also affected by the referential properties of these two elements and subsequently by their case marking. Non-specific and non-referential NPs tend to appear adjacent to the verb form, whereas specific NPs tend to appear in sentence initial position. Thus, in a situation where both S and O are specific (marked with nominative and accusative respectively) their relative order may be SO or OS (17a–17b). When both S and O are non-specific (both carrying no overt case marking) their relative order will be SO (17c). The same order occurs when S is specific and O is non-specific or non-referential (no overt case

marking) (17d). However, when O is specific (marked with accusative) and S is non-referential, the only available order is OS provided that the verb carries habitual aspect (17e–17f).

- (17) a. *both S and O are specific* → *SOV*:  
 adam bilet-i al-dı.  
 man ticket-ACC buy-PAST  
 ‘The man bought the ticket.’
- b. *both S and O are specific* → *OSV*:  
 bilet-i adam al-dı  
 ticket-ACC man buy-PAST  
 ‘The man bought the ticket.’
- c. *both S and O are non-specific* → *SOV*:  
 bir adam bir bilet al-dı.  
 a man a ticket buy-PAST  
 ‘A man bought a ticket.’
- d. *S is specific and O is non-specific or non-referential* → *SOV*:  
 adam bir bilet al-dı.  
 man a ticket buy-PAST  
 ‘The man bought a ticket.’
- e. *S is non-referential and O is specific* → *OSV*:  
 bilet-i adam al-ır.  
 ticket-ACC man buy-HAB  
 ‘Men buy the ticket.’
- f. *S is non-referential and O is specific* → *\*SOV*:  
 \*adam bilet-i al-dı  
 man ticket-ACC buy-PAST

### 3 Greek vs. Turkish

In the remainder of this section, we compare the case systems of Greek and Turkish and we discuss the interaction of specificity and word order in these languages.

The functions of nominative, genitive and accusative coincide to some extent in both languages. Specifically, nominative marks the subject in both languages (1 and 7); genitive denotes possession in both languages (3b and 9) but also indirect objects in Greek (3a); accusative is used for direct objects in both languages (2 and 8). Moreover, Greek expresses the functions of Turkish dative, ablative and locative by means of prepositional phrases (4–6).

However, accusative in Turkish is also a special specificity marker (see example 14 above). Specific objects are necessarily case marked (17a and 17b) and obligatorily appear in sentence initial positions when the subject is non-referential (17d vs. 17e).

In contrast, Greek does not exhibit any interaction between case and specificity, in the sense that there is no case alternation with respect to specificity. In fact, Greek does not mark specificity at all. Definiteness is marked by means of the definite article, whereas



indefiniteness is marked by the indefinite article or by the absence of article (Holton et al., 1997). Crucially, there is no way to distinguish between indefinite specific and indefinite non-specific noun phrases, since both are marked by the indefinite article and carry case marking:<sup>5</sup>

- (18) o           janis           ayorase           enan   pinaka.  
       the-NOM John-NOM buy-PAST.3SG a-ACC painting-ACC  
       ‘John bought some painting.’ or ‘John bought a certain painting.’

## IV Previous studies on the acquisition of case in Turkish as an L2

The acquisition of case by L2 learners of Turkish has been examined recently by Gürel (2000) and Haznedar (2006). The aim of both studies was to investigate whether the difficulties L2 learners of Turkish exhibit with respect to case morphology are due to the realization of case morphemes or alternatively whether they reflect a more global syntactic deficit.

Gürel (2000) conducted two tasks with 21 English-speaking learners of Turkish, who had been divided into three proficiency groups (7 participants in each group), and a group of 15 native speakers (NS) of Turkish. The first experiment was a grammaticality judgement task whereas the second one was a production task that involved picture description. The results showed that the L2 learners had an overall lower performance than the NS and that there were developmental patterns. More specifically, the intermediate and advanced learners performed significantly higher than the beginners. Moreover, the learners produced omissions rather than substitutions of case morphemes and displayed sensitivity to the word order constraints of Turkish. In particular, Gürel tested sentences with (non)specific (non)adjacent objects and found that L2 participants did not accept the ungrammatical non-specific non-adjacent items at a high rate. These findings indicate knowledge of word order restrictions in the L2 grammar. Thus, Gürel argues for a deficit at the syntax-morphology interface rather than the syntactic module *per se*.

Haznedar (2006) draws similar conclusions based on spontaneous production data (6 recordings) collected from an English-speaking learner of Turkish during the participant's 5-month stay at Boğaziçi University. The findings showed that the learner was able to use verbal suffixes correctly. More specifically, he was more than 90% correct in his production of tense marking. He also showed evidence for use of all S–V agreement morphemes and he marked subjects correctly for nominative. In contrast, the L2 learner exhibited very low performance in case marking (accusative, dative, locative and ablative). However, he seemed to be aware of word order restrictions in Turkish, since he was consistent in the correct case marking of scrambled objects. On the basis of these findings, Haznedar concludes that the learner faced difficulties only with the surface realization of Case.

In what follows we present our study with adult Greek learners of Turkish in an attempt to contribute to the on-going debate regarding the source of optionality in L2 grammars. Taking into account the grammatical properties of Greek and Turkish, as well as the theories of L2 acquisition presented in Section II, we make the following predictions:

- The organic grammar account (Vainikka and Young-Scholten, 1994, 1996, 2006, 2007, 2009) predicts that functional categories are missing from early L2 grammars and that Greek L2 learners of Turkish at the beginners' level will have lexical VPs and NPs that reflect the L1 grammar, i.e. Greek. Within this account, we expect that we will find no evidence for the presence of functional categories related to Tense, Agreement and Definiteness. Consequently, no use or very low performance on verbal and nominal suffixes is expected to occur at that level. Moreover, both SOV and OSV orders in Turkish are expected to be problematic for Greek learners, given that the object in Greek is not base-generated in pre-verbal positions (Philippaki-Warbuton, 1985; Tsimpli, 1990; Holton et al., 1997; Alexiadou and Anagnostopoulou, 1998; Keller and Alexopoulou, 2001; Clairis and Babinotis, 2005; Lascaratou and Georgiafentis, 2006) and that the SOV and OSV orders, although possible, are marginal (Keller and Alexopoulou, 2001; Lascaratou and Georgiafentis, 2006: 18–19). In addition, due to the lack of functional categories there will be no sensitivity to the interaction between word order restrictions and case morphology.
- The Full Transfer/Full Access account and the so-called Morphological Deficit account predict transfer of the relevant functional categories (TP, AgrP, DP) from the L1 and, hence, production of both verbal and nominal morphemes. However, variability in the use of morphology is expected due to mapping problems between the abstract syntactic features and their morphological realization. Furthermore, the Full Transfer/Full Access account predicts that any initial difficulties with the SOV and OSV orders will be rapidly overcome due to the fact that they are the outcome of parameter (re)setting. Under this account, although sensitivity to the interaction between case suffixation and word order restrictions may not be evidenced in the initial stage, it is expected that sensitivity will appear quite early given the presence of functional categories and the lack of representational deficits in L2 grammars.
- Although the Failed Functional Features and the Interpretability accounts make predictions for the final stage of L2 acquisition, we will assume that their claims may be relevant to L2 development. Thus, features that are grammaticalized in the L1 (Agreement, Tense and Case) are not expected to cause difficulties for L2 learners. With respect to the SOV and OSV orders, the Failed Functional Features and the Interpretability accounts make the same predictions with the Full Transfer/Full Access theory. Finally, these accounts predict some sensitivity to the word order constraints in L2 Turkish, because the relevant features, i.e. referentiality and specificity,<sup>6</sup> even though not grammaticalized in Greek, are interpretable.

## V Method

### 1 Participants

All the non-native speakers of Turkish who participated in this study were monolingual native speakers of Greek. Moreover, they all learnt Turkish as a foreign language at the Department of Mediterranean Studies of the University of the Aegean through instruction. None of them had spent more than six months in a Turkish-speaking environment when tested. The L2 learners were divided into three proficiency levels based on whether at the time of testing they were attending the first, second or third level of studies on Turkish at

the University. More specifically, the beginners (first level) have attended Turkish lessons for two years, the low intermediates (second level) for three years and the high intermediates (third level) for four years. The duration of the Turkish language course in each proficiency level is 24 weeks, six hours per week. Although attendance is not obligatory, students attend classes regularly in order to avoid failure in the exams, a fact confirmed by their instructors.

The cloze task has been conducted with 35 beginner, 37 low intermediate and 39 high intermediate learners of Turkish. In the sentence-picture matching task, there were 48 Greek learners of Turkish: 9, 20 and 19 participants in the first, second and third proficiency level respectively. The on-line grammaticality judgement task has been carried out with fifty-six learners (14 beginners, 14 low intermediates and 28 high intermediates). All three tasks have also been administered to 16 native speakers of Turkish who either lived in Turkey or immigrated in Greece in their adulthood (13 females and 3 males; mean age: 29.75 years). None of them were Turkish-Greek early bilinguals.

## 2 Cloze task

A cloze task was used to examine the L2 learners' overall performance on verbal and nominal suffixes, as it constitutes a speedy and efficient way to assess non-native speakers' production of grammatical morphemes. The cloze task consisted of a text created by two native speakers of Turkish and it contained 61 gaps. All the gaps involved nominal (30 gaps) and verbal (31 gaps) morphemes denoting case, number, tense and subject-verb agreement. The tenses tested were present (seven gaps), perfective past (four gaps) and future (five gaps). The present and the future share the same paradigm of subject-verb suffixes, whereas the subject-verb suffixes for the perfective past are different. In the perfective past, we tested the 1st singular (three gaps) and the 3rd plural (one gap). In the present and the future, the 1st singular (six gaps) and the 3rd singular (four gaps) were examined. Moreover, there was one gap that required the addition of the infinitival suffix.

All the nominal morphemes included in the cloze task involved case suffixes. In this article, we focus on the findings that relate to the morphological marking of specific and non-specific direct objects. The accusative case morpheme was required for all specific direct objects of the task (four gaps), whereas the non-specific objects should be left unsuffixed (two gaps).<sup>7</sup> The other case suffixes tested were distributed as follows: eleven gaps for the dative case, five gaps for the ablative case, four gaps for the genitive case and four gaps for the locative case.

## 3 Sentence picture matching (SPM) task

A sentence picture matching task was used to investigate whether the L2 learners make use of case morphology and word order to comprehend the argument structure of simple transitive sentences.

*a Materials:* The SPM task contained 36 items. In this article we concentrate on the set of the 24 sentences, which were designed to test the interaction of word order and case morphology. More specifically, the critical sentences the participants heard belonged to three conditions:

- *SO<sub>acc</sub>V: specific object (marked for accusative case):*  
(19) erkek kadın-ı kucakl-ıyor.  
man woman-ACC embrace-PRES  
'The man is embracing the woman.'
- *O<sub>acc</sub>SV: specific object (marked for accusative case):*  
(20) kadın-ı erkek kucakl-ıyor.  
woman-ACC man embrace-PRES  
'It is a man that is embracing the woman.'
- *SO<sub>abs</sub>V: non-specific object (unmarked for case):*  
(21) bir erkek bir kaplan öl-dür-üyor.  
a man a tiger die-CAUSE-PRES  
'A man kills a tiger.'

**b Procedure:** The experiment took place in a quiet room, where the participants were individually tested by the first author. The participants were given a booklet that contained 36 quartets of pictures (see Appendix 1 for a sample). In each quartet, one picture was the target, whereas another picture depicted the same action with reversed thematic roles, i.e. for (19) and (20) a woman embracing a man. The other two pictures in each quartet described a different action (i.e. for (19) and (20) the action of greeting) performed by the same protagonists, in one of which the thematic roles were as in the target picture (a man greeting a woman), whereas in the other one the thematic roles were reversed (a woman greeting a man). The participants looked carefully at each quarter and at the same time they listened to a Turkish sentence. They were instructed to indicate the picture that best illustrated the sentence they heard. All 36 sentences were spoken by a female native speaker of Turkish and were tape-recorded.

#### 4 On-line grammaticality judgement (GJ) task

An on-line grammaticality judgement task was employed to investigate whether the L2 learners are able to use case morphology and word order constraints to judge the grammaticality of simple transitive sentences. Moreover, reaction times were measured in order to examine whether the native and non-native speakers differ in terms of the time required to make these judgements.

**a Materials:** The on-line GJ task consisted of 48 sentences: 24 experimental items and 24 pseudo-distractors.<sup>8</sup> The 24 experimental items were equally distributed across four conditions (6 items per condition), as shown below (see Appendix 2 for the entire list of the experimental sentences):

- *SO<sub>acc</sub>V: specific object (marked for accusative case):*  
(22) çocuk gazete-ler-i oku-du.  
child newspaper-PL-ACC read-PAST  
'The child read the newspapers.'

- *O<sub>ACC</sub>SV*: specific object (marked for accusative case):  
 (23) gazete-ler-i çocuk oku-du.  
 newspaper-PL-ACC child read-PAST  
 'The newspapers were read by some child.'
- *SO<sub>ABS</sub>V*: non-specific/non-referential object (unmarked for case):  
 (24) çocuk gazete oku-du.  
 child newspaper read-PAST  
 'The child read some newspaper/newspapers.'
- \**O<sub>ABS</sub>SV*: non-referential object (unmarked for case)  
 (25) \*gazete çocuk oku-du  
 newspaper child read-PAST

The 24 experimental items consisted of 18 grammatical and 6 ungrammatical sentences.

In addition, 6 out of the 24 pseudo-distractors were grammatical and 18 were ungrammatical. The pseudo-distractors were of the same length as the experimental items; they all contained 3 words. The grammatical pseudo-distractors contained predication constructions, negative sentences as well as sentences consisting of a subject, a PP and a verb. The ungrammatical pseudo-distractors involved S-V agreement and tense mismatches (nine sentences) and incorrect morphological marking for cases other than accusative (nine sentences).

**b Procedure:** The experiment was designed through the e-prime paradigm (Schneider et al., 2002). The participants read the sentences word-by-word on a computer screen. Each word automatically appeared in the middle of the screen and remained visible for 2 seconds. At the end of each sentence a question mark appeared '?' which indicated that at this point the participant had to judge the grammaticality of the sentence. The participants had up to 4 seconds after they read the final word of each sentence to make their judgements. They had to press a green button for 'yes' responses and a red button for 'no' responses. Both buttons were specified on the keyboard. If no response was given after 4 seconds, the experiment continued with the presentation of the following sentence. Before the main experimental session, there was a familiarization phase consisting of 10 trials.

## VI Results

### 1 Cloze task

First, we present the accuracy on the verbal suffixes per proficiency group. The NS were 100% accurate in all their responses. The data in Table 1 illustrate developmental differences among the three L2 groups in both the Agreement and Tense suffixes. More specifically, the low intermediate group performs better than the beginners (Agreement:  $\chi^2 = 109.436$ ,  $p < 0.001$ ; Tense:  $\chi^2 = 91.416$ ,  $p < 0.001$ ) and worse than the high intermediate group (Agreement:  $\chi^2 = 108.743$ ,  $p < 0.001$ ; Tense:  $\chi^2 = 63.676$ ,  $p < 0.001$ ).

The following table presents the percentages of correct productions for the cases tested per each proficiency level. The data shown in Table 2 indicate a clear developmental pattern: in all conditions, apart from the non-specific object one, the low intermediate

**Table 1** Verbal suffixes: Correct scores per proficiency level

| Verbal suffixes | Level I (N = 35) | Level II (N = 37) | Level III (N = 39) |
|-----------------|------------------|-------------------|--------------------|
| S-V agreement   | 42% (220/525)    | 73% (407/555)     | 95% (559/585)      |
| Tense           | 49% (272/560)    | 76% (449/592)     | 92% (577/624)      |
| Total           | 45% (492/1085)   | 75% (856/1147)    | 94% (1136/1209)    |

**Table 2** Case suffixes: Correct scores per proficiency level

| Cases                          | Level I (N = 35) | Level II (N = 37) | Level III (N = 39) |
|--------------------------------|------------------|-------------------|--------------------|
| Specific object (accusative)   | 21% (29/140)     | 39% (58/148)      | 49% (77/156)       |
| Non-specific object (unmarked) | 76% (53/70)      | 64% (47/74)       | 62% (48/78)        |
| Other cases                    | 28% (253/910)    | 41% (393/962)     | 58% (588/1014)     |
| Total                          | 30% (335/1120)   | 42% (498/1184)    | 57% (713/1248)     |

students performed better than the beginners, and the high intermediate students produced more correct responses than the low intermediate learners.

In the other cases, all L2 groups differed significantly from each other (beginners vs. low:  $\chi^2 = 35.240$ ,  $p < 0.001$ ; beginners vs. high intermediates:  $\chi^2 = 177.611$ ,  $p < 0.001$ ; low vs. high intermediates:  $\chi^2 = 57.985$ ,  $p < 0.001$ ).

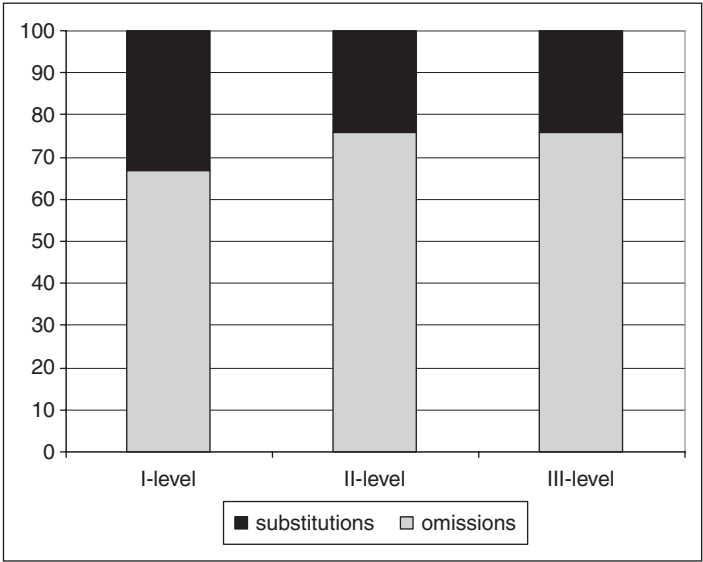
In the definite object condition, the beginners performed significantly worse than the low ( $\chi^2 = 11.647$ ,  $p < 0.01$ ) and high intermediate ( $\chi^2 = 26.337$ ,  $p < 0.001$ ) learners, whereas the difference between the low and high intermediates was not significant ( $p > 0.05$ ).

When the direct object was non-specific and thus the noun was morphologically unmarked, there were no statistically significant differences among the three groups ( $p > 0.05$ ). Notice that in this condition the beginners seem to perform better – even though not significantly – than the intermediate groups. However, this is probably a misleading result because in this case correct performance coincides with omission and, hence, we cannot be sure whether the production of a null morpheme really reflects knowledge of the constraints related to indefinite objects in Turkish (see fn 4).

Moreover, we counted the omission and substitution errors produced by each proficiency group only for the definite-object condition. Figure 1 illustrates that the omission errors are more frequent than the substitution errors in all proficiency groups and the statistical analyses showed that these differences are significant (beginners:  $\chi^2 = 12.333$ ,  $p < 0.001$ ; low intermediates:  $\chi^2 = 23.511$ ,  $p < 0.001$ ; high intermediates:  $\chi^2 = 21.278$ ,  $p < 0.001$ ).

We have also conducted a more detailed error analysis in order to see which case suffixes have been mistakenly used instead of the correct ones in the definite and indefinite object conditions. All responses that did not involve case suffixes have been excluded from this analysis. Table 3 presents the erroneous case suffixes produced for the definite object condition. As shown in Table 3, the suffix that is most often erroneously supplied by all L2 groups is the one marked for dative.

Table 4 presents the analysis of substitution errors for the indefinite object condition. The data show that the beginners erroneously use accusative, dative and locative suffixes



**Figure 1** Specific object: Omission and substitution errors per proficiency group (%)

**Table 3** Specific object: Erroneous case suffixation (%)

| Proficiency levels | Dative      | Locative    | Ablative    | Genitive    |
|--------------------|-------------|-------------|-------------|-------------|
| I (N = 35)         | 63% (20/32) | 37% (12/32) | 0           | 0           |
| II (N = 37)        | 58% (7/12)  | 25% (3/12)  | 8.5% (1/12) | 8.5% (1/12) |
| III (N = 39)       | 70% (7/10)  | 30% (3/10)  | 0           | 0           |

**Table 4** Non-specific object: Error analysis

| Proficiency level | Accusative    | Dative        | Locative      | Ablative | Genitive  |
|-------------------|---------------|---------------|---------------|----------|-----------|
| I (N = 35)        | 33.33% (4/12) | 33.33% (4/12) | 33.33% (4/12) | 0        | 0         |
| II (N = 37)       | 84% (10/12)   | 8% (1/12)     | 8% (1/12)     | 0        | 0         |
| III (N = 39)      | 90% (19/21)   | 5% (1/21)     | 0             | 0        | 5% (1/21) |

to the same degree, whereas both intermediate groups over-apply the accusative suffix more often than other case suffixes.

2 Sentence-picture matching task

We present the overall accuracy scores per proficiency group in Table 5. The overall scores show developmental patterns in that the high ( $\chi^2 = 42.449, p < 0.001$ ) and low ( $\chi^2 = 13.530, p < 0.001$ ) intermediate groups perform significantly better than the beginners, and the beginners significantly worse than the low intermediates ( $\chi^2 = 14.009, p < 0.001$ ).



**Table 5** SPM task: Overall accuracy scores

| Proficiency level | Correct responses |
|-------------------|-------------------|
| I (N = 9)         | 28% (103/378)     |
| II (N = 20)       | 38% (320/840)     |
| III (N = 19)      | 47% (377/798)     |

**Table 6** SPM task: Accuracy scores for the experimental items

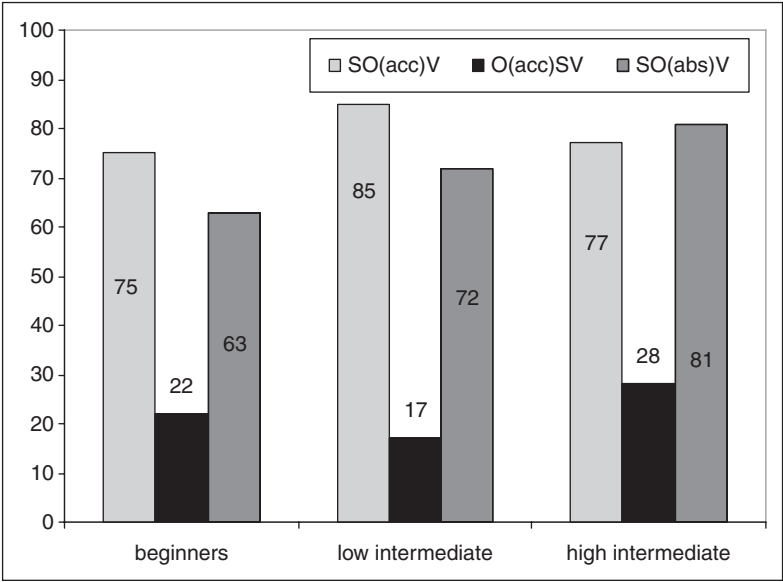
| Proficiency level | SO <sub>ACC</sub> V | O <sub>ACC</sub> SV | SO <sub>ABS</sub> V |
|-------------------|---------------------|---------------------|---------------------|
| I (N = 9)         | 35% (17/48)         | 8% (4/50)           | 35% (32/92)         |
| II (N = 20)       | 48% (58/120)        | 12% (14/119)        | 50% (119/238)       |
| III (N = 19)      | 55% (61/110)        | 19% (22/113)        | 57% (125/220)       |

Table 6 illustrates the frequency of correct responses for the experimental items. The NS were 100% accurate in all target sentences.<sup>9</sup> The high intermediate group’s performance was significantly better than that of the beginners in both the SO<sub>ACC</sub> V ( $\chi^2 = 5.368$ ,  $p < 0.03$ ) and the SO<sub>ABS</sub> V ( $\chi^2 = 12.600$ ,  $p < 0.001$ ) conditions, whereas the low intermediates differed from the beginners only in the SO<sub>ABS</sub> V condition ( $\chi^2 = 6.191$ ,  $p < 0.02$ ). No significant differences among the three groups were observed in the O<sub>ACC</sub> SV condition. Notice, however, that most of the developmental differences in the SO<sub>ACC</sub> V and SO<sub>ABS</sub> V sentences between the low or high intermediates on the one hand and the beginners on the other hand are probably due to lexical and not grammatical errors. This is obvious in Figure 2, which represents the accuracy in percentages per each group and condition when only grammatical – in this case word order – errors are counted. The only difference that remains statistically significant is the one between the beginners and the high intermediates in the SO<sub>ABS</sub> V condition ( $\chi^2 = 8.036$ ,  $p < 0.01$ ).

3 On-line grammaticality judgement task

*a Accuracy for pseudo-distractors:* Table 7 shows the accuracy scores for the grammatical and ungrammatical pseudo-distractors for the L2 learners and the NS. An ANOVA with Grammaticality (grammatical vs. ungrammatical sentences) as the within-participants factor and Group (native speakers vs. beginners vs. low vs. high intermediates) as the between-participants factor revealed a significant main effect of Grammaticality ( $F(1,68) = 20.392$ ,  $p < 0.001$ ,  $\eta^2 = 0.231$ ) due to the fact that overall the grammatical sentences have been judged significantly more accurately than the ungrammatical ones and a significant Group effect ( $F(3,68) = 23.275$ ,  $p < 0.001$ ,  $\eta^2 = 0.507$ ). Planned post-hoc comparisons showed that the Group effect is due to the fact that (a) the native speakers performed significantly better than all L2 groups ( $p < 0.001$ ) and (b) the beginners performed significantly worse than the high intermediate group ( $p < 0.03$ ). One-way





**Figure 2** SPM task:Accuracy scores per condition and proficiency group (%)

**Table 7** GJ task:Accuracy scores for the pseudo-distractors (percentages)

| Groups                   | Grammatical pseudo-distractors | Ungrammatical pseudo-distractors |
|--------------------------|--------------------------------|----------------------------------|
| I (N = 14)               | 54                             | 37                               |
| II (N = 14)              | 59                             | 43                               |
| III (N = 28)             | 71                             | 49                               |
| Native speakers (N = 16) | 89                             | 87                               |

ANOVAs performed on each proficiency level showed that the grammatical sentences were judged significantly better than the ungrammatical ones only by the low ( $F(1, 26) = 6.782, p < 0.02$ ) and high ( $F(1,55) = 19.383, p < 0.001$ ) intermediate groups.

Furthermore, we performed separate analyses on the ungrammatical pseudo-distractors to explore whether the L2 learners' low accuracy scores were attested in a specific type of ungrammaticality.<sup>10</sup> Our ungrammatical pseudo-distractors involved incorrect use of verbal suffixes (subject–verb agreement and tense) and case suffixes. The accuracy percentages per structure and group are presented in Table 8. A two-way ANOVA with Structure as the within-participants factor and Group as the between-participants factor revealed a significant main effect for Structure ( $F(1,2,67) = 28.589, p < 0.001, \eta^2 = 0.460$ ) and Group ( $F(1,3,68) = 20.526, p < 0.001, \eta^2 = 0.475$ ), as well as a significant interaction between Structure and Group ( $F(1,3,68) = 9.233, p < 0.001, \eta^2 = 0.289$ ). The Bonferroni post-hoc comparisons showed that all L2 groups performed significantly worse than the NS ( $p < 0.001$ ), whereas the L2 groups did not

**Table 8** GJ task:Accuracy scores per ungrammatical structure (%)

| Groups                   | (*) Verbal<br>suffixes | (*) Case<br>suffixes |
|--------------------------|------------------------|----------------------|
| I (N = 14)               | 48                     | 27                   |
| II (N = 14)              | 60                     | 27                   |
| III (N = 28)             | 77                     | 34                   |
| Native speakers (N = 16) | 88                     | 88                   |

**Table 9** GJ task:Accuracy scores for the experimental items (%)

| Groups                   | (*)Object <sub>ABS</sub> | Object <sub>ABS</sub> | Object <sub>ACC</sub> | Object <sub>ACC</sub> |
|--------------------------|--------------------------|-----------------------|-----------------------|-----------------------|
|                          | OSV                      | SOV                   | OSV                   | SOV                   |
| I (N = 14)               | 41                       | 62                    | 51                    | 67                    |
| II (N = 14)              | 42                       | 64                    | 56                    | 76                    |
| III (N = 28)             | 31                       | 67                    | 61                    | 74                    |
| Native speakers (N = 16) | 86                       | 86                    | 83                    | 91                    |

significantly differ from each other ( $p > 0.1$ ). Furthermore, one-way ANOVAs performed on each ungrammatical structure revealed significant group effects (Verbal morphemes:  $F(3,68) = 3.902$ ,  $p < 0.05$ ; Case:  $F(3,68) = 40.034$ ,  $p < 0.001$ ). Nevertheless, the Bonferroni post-hoc comparisons illustrated that the NS significantly differed from all L2 groups in the ungrammatical sentences involving erroneous case suffixes ( $p < 0.001$ ), whereas they were significantly different only from the beginners in the sentences involving erroneous verbal suffixes. This finding corroborates the discrepancy between the nominal and the verbal domain also attested in the data from the cloze task.

*b Accuracy for experimental items:* A repeated-measures ANOVA with Word Order (canonical vs. non-canonical sentences) and Case (accusative vs. absolutive) as the within-participants factors and Group (native speakers vs. beginners vs. low vs. high intermediates) as the between-participants variable has been performed. The Group effect was significant ( $F(3,68) = 29.473$ ,  $p < 0.001$ ,  $\eta^2 = 0.565$ ;  $F(3,92) = 30.208$ ,  $p < 0.001$ ,  $\eta^2 = 0.496$ ) and planned post-hoc comparisons with the Bonferroni test showed that this is due to the significant comparisons between the NS and all L2 groups ( $p < 0.001$  in the subject- and the item-analyses), whereas there were no significant differences among the L2 groups ( $p = 1.000$  in the subject- and the item-analyses). Moreover, the main effects of Case ( $F(1,68) = 13.603$ ,  $p < 0.001$ ,  $\eta^2 = 0.167$ ;  $F(1,92) = 17.275$ ,  $p < 0.001$ ,  $\eta^2 = 0.158$ ) and Word Order ( $F(1,68) = 57.353$ ,  $p < 0.001$ ,  $\eta^2 = 0.458$ ;  $F(1,92) = 46.191$ ,  $p < 0.001$ ,  $\eta^2 = 0.334$ ) as well as the interaction between Word Order and Group ( $F(3,68) = 4.582$ ,  $p < 0.01$ ,  $\eta^2 = 0.168$ ;  $F(3,92) = 5.470$ ,  $p < 0.01$ ,  $\eta^2 = 0.151$ ) was significant. To further explore the source of this interaction a two-way repeated measures ANOVA with Case

**Table 10** GJ task: RTs per condition and group for experimental sentences

| Groups                   | (*)Object <sub>ABS</sub><br>OSV | Object <sub>ABS</sub><br>SOV | Object <sub>ACC</sub><br>OSV | Object <sub>ACC</sub><br>SOV |
|--------------------------|---------------------------------|------------------------------|------------------------------|------------------------------|
| I (N = 14)               | 1219                            | 1003                         | 862                          | 1017                         |
| II (N = 14)              | 992                             | 798                          | 819                          | 851                          |
| III (N = 28)             | 1459                            | 1021                         | 1025                         | 1034                         |
| Native speakers (N = 16) | 975                             | 1090                         | 1164                         | 897                          |

and Word order as the within-participants factors has been performed on the data from each participant group. In this and all further analyses, the L2 data from the three groups have been collapsed together, since the planned comparisons did not reveal any significant differences among the L2 groups.

The L2 group manifested significant Word Order ( $F(1,55) = 74.094$ ,  $p < 0.001$ ,  $\eta^2 = 0.574$ ;  $F(1,71) = 50.435$ ,  $p < 0.001$ ,  $\eta^2 = 0.415$ ) and Case effects ( $F(1,55) = 20.755$ ,  $p < 0.001$ ,  $\eta^2 = 0.274$ ;  $F(1,71) = 15.495$ ,  $p < 0.001$ ,  $\eta^2 = 0.179$ ) as well as a significant interaction between Case and Word Order ( $F(1,55) = 5.426$ ,  $p < 0.03$ ,  $\eta^2 = 0.090$ ;  $F(1,71) = 5.517$ ,  $p < 0.03$ ,  $\eta^2 = 0.072$ ). These effects are due to:

- more accurate judgements for sentences with canonical than with non-canonical word order (accusative ( $t(55) = 5.195$ ,  $p < 0.001$ ;  $t(71) = 3.144$ ,  $p < 0.01$ ); absolutive: ( $t(55) = 6.287$ ,  $p < 0.001$ ;  $t(71) = 6.917$ ,  $p < 0.001$ ))
- more accurate judgements for sentences involving the accusative than the absolutive case (SOV:  $t(55) = 2.064$ ,  $p < 0.05$ ; OSV:  $t(55) = 4.516$ ,  $p < 0.05$ ;  $t(71) = 4.419$ ,  $p < 0.001$ ).

On the other hand, the data from the control group did not reveal any statistically significant effects.

*c Reaction time data:* We turn now to the reaction time (RT) data. Table 10 presents the RTs for the correct trials. The data in Table 10 show that in the sentences involving the accusative case, the NS but not the L2 learners were slower in the items with non-canonical than with canonical word order. Moreover, in the absolutive case the NS' behaviour was different from the L2 learners'. Specifically, the ungrammatical condition, i.e. O<sub>ABS</sub>SV, was the slowest and, hence, the most difficult condition in the L2 data, whereas this was not the case for the NS. These observations have been confirmed by statistical analyses.

In the repeated-measures ANOVA with Case and Word Order as the within-participants factors and Group as the between-participants factor, the Group effect was significant only in the item-analysis ( $F(3,71) = 3.190$ ,  $p < 0.03$ ,  $\eta^2 = 0.119$ ). Moreover, the Bonferroni post-hoc comparisons showed a significant difference between the low and the high intermediate groups, which was significant only in the item analysis ( $p < 0.03$ ). On the contrary, the main effects of Case ( $F(1,63) = 5.289$ ,  $p < 0.03$ ,  $\eta^2 = 0.077$ ;  $F(1,71) = 5.288$ ,  $p < 0.03$ ,  $\eta^2 = 0.069$ ) and Word Order ( $F(1,63) = 6.848$ ,  $p < 0.02$ ,  $\eta^2 = 0.098$ ;  $F(1,71) = 7.020$ ,

$p < 0.02$ ,  $\eta^2 = 0.090$ ) as well as the three-way interaction among Case, Word Order and Group ( $F(1,63) = 3.951$ ,  $p < 0.02$ ,  $\eta^2 = 0.158$ ;  $F(3,71) = 4.515$ ,  $p < 0.01$ ,  $\eta^2 = 0.160$ ) were significant. To further explore the three-way interaction, two-way ANOVAs have been performed for each group data set. As for the accuracy data, the RTs from the three L2 groups have been collapsed together as no significant differences were revealed.

The NS data revealed a significant interaction between Case and Word Order ( $F(1,15) = 6.102$ ,  $p < 0.03$ ,  $\eta^2 = 0.289$ ;  $F(1,23) = 11.410$ ,  $p < 0.01$ ,  $\eta^2 = 0.332$ ). Paired-samples  $t$ -tests showed that this interaction is due to the faster RTs for (a) the SOV than the OSV sentences in the accusative case condition ( $t(23) = 3.780$ ,  $p < 0.01$ ) and (b) the OSV sentences in the absolute than the accusative case ( $t(23) = 2.504$ ,  $p < 0.03$ ).

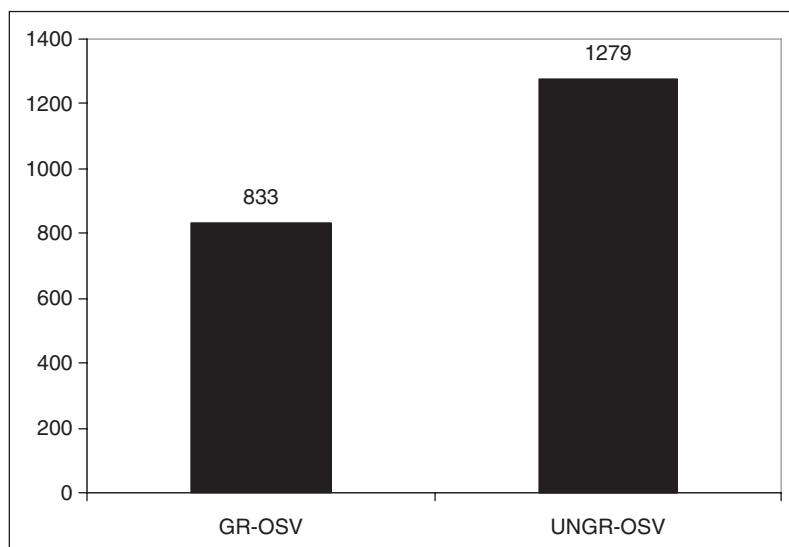
In the L2 data, the two-way ANOVA with case and order as the within-participants factors revealed that the main effects of Case ( $F(1,50) = 7.896$ ,  $p < 0.01$ ,  $\eta^2 = 0.136$ ;  $F(2,150) = 6.366$ ,  $p < 0.02$ ,  $\eta^2 = 0.113$ ) and Word Order ( $F(1,50) = 9.338$ ,  $p < 0.01$ ,  $\eta^2 = 0.157$ ;  $F(2,150) = 5.344$ ,  $p < 0.03$ ,  $\eta^2 = 0.097$ ) as well as the interaction between Case and Word Order ( $F(1,50) = 10.415$ ,  $p < 0.01$ ,  $\eta^2 = 0.172$ ;  $F(2,150) = 8.373$ ,  $p < 0.01$ ,  $\eta^2 = 0.143$ ) were significant. These results are due to the fact that the OSV condition in the absolute case was judged significantly slower than the SOV condition in the same case ( $t(51) = 4.017$ ,  $p < 0.001$ ;  $t(56) = 3.405$ ,  $p < 0.01$ ) and the OSV sentences in the accusative case ( $t(50) = 3.540$ ,  $p < 0.01$ ;  $t(56) = 3.670$ ,  $p < 0.01$ ).

Furthermore, we did an additional analysis on the items involving the absolute case. More specifically, we counted the RTs for the trials that have been erroneously judged as grammatical in the ungrammatical condition  $O_{ABS}SV$ . Figure 3 displays the mean RTs for all three L2 groups for the  $O_{ABS}SV$  trials judged as grammatical and ungrammatical. The figure shows a different pattern for the  $O_{ABS}SV$  trials judged as grammatical and ungrammatical: the RTs for the former are shorter than for the latter ( $t(50) = 5.383$ ,  $p < 0.001$ ;  $t(55) = 3.819$ ,  $p < 0.001$ ).

## VII Summary and discussion

Before we proceed to the interpretation of the findings discussed above, we will first summarize the main results of the three experimental tasks employed in our study.

The results of the cloze task clearly show developmental effects in the production of all case suffixes. With respect to the production of the correct accusative morphemes in the specific object condition, the two intermediate groups performed significantly better than the beginners. In this condition, omission errors significantly outnumbered substitution errors in all proficiency levels and dative suffixes were overused more frequently than other case suffixes. As far as the non-specific object condition is concerned, the beginners performed better than the intermediate groups, even though the differences did not reach significance. It is possible that this finding is an artefact of the way the cloze task was designed and executed, since in the case of indefinite objects correct performance means production of a null morpheme. Notice that the analysis of the substitution errors in the non-specific object condition provides evidence that the intermediate groups are actually at a higher developmental stage compared to the beginners. This is because the intermediate groups overgeneralize the accusative suffix more often than the other case suffixes, a finding which is not observed in the beginners.



**Figure 3**  $O_{ABS}SV$  conditions: Comparison of trials erroneously judged as grammatical and trials correctly judged as ungrammatical

Interestingly, the SPM task yielded an overall better performance on sentences with SOV word order than on sentences with OSV word order. This is supported by the performance of all proficiency groups. Attested differences among them are attributed to the fact that L2 speakers occasionally confused the meaning of the verbs used in the sentences.

Finally, the on-line GJ task indicates that the result pattern of the L2 learners differs from that of the NS although no significant differences among the three proficiency groups are observed. More specifically, the NS were found to be more accurate than the L2 learners. In addition, the L2 learners exhibited difficulties in correctly judging (a) the grammatical sentences with  $O_{ACC}SV$  word order and (b) the ungrammatical  $O_{ABS}SV$  sentences, which was not the case for the NS. The RT data also demonstrate significant differences between the NS and the L2 learners, in that the ungrammatical  $O_{ABS}SV$  condition was the slowest condition in the L2 but not in the NS data. Finally, sentences with non-canonical word order resulted in slower RTs than those with canonical word order, even though this effect did not yield any significant differences in the L2 learners' data.

Our data from the cloze task reveal a difference in performance between the verbal and the nominal suffixes in all proficiency levels, with better performance for the verbal morphemes. More specifically, the intermediate groups exhibited a rather high accuracy with respect to S–V agreement (73% for the low and 95% for the high intermediates) and Tense (76% for the low and 92% for the high intermediates). This result contrasts with the low accuracy in the production of Case (accusative: 21% for the beginners, 39% for the low intermediates and 49% for the high intermediates; other cases: 28% for the beginners, 41% for the low intermediates and 58% for the high intermediates). Our findings are consistent with Haznedar's (2006) study, which clearly suggests an asymmetry in the use of verbal and case morphology.

Interestingly, the difference between the verbal and the nominal suffixes is not predicted by any of the L2 theories. Under the organic grammar account, no verbal suffixes are expected in the early L2 stages, given the unavailability of the corresponding functional categories. Despite the fact that both verbal and nominal suffixes are more erroneously used by the beginners than the intermediate groups, only the former seem to have been acquired at least by the high intermediate group, given that their accuracy is above 90%. Nonetheless, one could argue that the low accuracy score (45%) for the verbal suffixes observed in the beginners may be an indication of lack of the functional categories of Agr and Tense. However, it is not clear how even this accuracy percentage is achieved under the assumption that no functional projections are expected to exist at early stages of L2 acquisition.

Turning to the morphological deficit theory, there are no grounds for assuming that the nominal suffixes will be harder to acquire than the verbal ones, since all functional categories should be available and mapping problems should affect equally the verbal and the nominal domain. Similarly, the Failed Functional Features (Hawkins and Chan, 1997; Hawkins and Hattori, 2006) and the Interpretability hypotheses (Tsimplici, 2003; Tsimplici and Dimitrakopoulou, 2007) cannot account for the discrepancy between the verbal and the nominal domain, since all the features involved are grammaticalized in the L1. Even though we cannot make any claims about the final stages of acquisition of the language under investigation we strongly believe that the divergence in the development of verbal and nominal features deserves further attention within these L2 theories. One possible scenario is that the richness of the L2 morphological paradigm as well as the way functional features are realized in the L1 and the L2 affect the learners' performance. For instance, Turkish has five morphologically marked cases, whereas Greek has only three. Hence, the difference between the two case systems and the morphological realization of the cases in the two languages may be the reason for the lower performance of Greek learners of Turkish with respect to the nominal system. This suggests that the difficulties observed in the L2 development may not be entirely dependent on the interpretability or grammaticalization of the relevant features but also on the differences between the two grammatical systems which are attested at the pure morphological level.

Moreover, the results of the cloze task replicate the optionality observed by previous studies in L2 Turkish with respect to the production of case suffixes (Gürel, 2000; Haznedar, 2006). In agreement with other studies (Gürel, 2000; Prévost and White, 2000), we found more omissions than substitutions, which suggests, despite the variable use of case morphology, that L2 learners prefer to avoid the production of erroneous suffixes. This has been taken as an indication of some internal knowledge on the part of the learners (cf. Borer and Rohrbacher, 1997; Gürel, 2000) and, thus, provides support to the Morphological Deficit hypothesis. In our study, this is corroborated by the error analysis of the non-specific object condition, in which the intermediate groups erroneously overuse the accusative suffix showing some knowledge of the associated functional projections (AgrOP). In contrast, the beginners overuse not only the accusative but also the dative and the locative suffixes. However, one still needs to provide an explanation for the 30% of substitutions observed in our data<sup>11</sup> as well as for the low performance on case suffixes even by the high intermediate learners, which obviously cannot be easily accounted for by the Morphological Deficit account.

The data from the SPM and the on-line GJ tasks show that the learners were highly accurate in accepting SOV word orders as grammatical. This result indicates that the learners have some knowledge of the word order patterns in Turkish. Significantly, it militates against the organic grammar account, which predicts transfer of the L1 word orders in the early interlanguage stages. The rather high performance on the SOV word order may be related to the fact that the SOV order is intimately linked to the head-final parameter. It is possible that a small amount of exposure to the language is required in order for the correct value of the parameter to be set. This explanation is in favour of the Full Transfer/Full Access, the Morphological Deficit, the Failed Functional Features and the Interpretability theories, which assume access to the principles and parameters of UG and rapid specification of the corresponding values according to the input. Nonetheless, based on the present data, we cannot claim with certainty that the parameter has been fully acquired. First, we do not have any production data on the particular phenomenon and, more importantly, we lack data on other aspects of the parameter, such as postpositions and the position of complementizers.

On the contrary, our data reveal a significant difference in performance between SOV and OSV sentences since the learners exhibited considerable difficulties with OSV word orders in both the SPM and the on-line tasks. This finding indicates that the learners do not allow the scrambling of the direct object. One way to interpret this result is to attribute it to a general processing strategy, according to which L2 learners (as well as native speakers) prefer to construe the first NP as the subject of the sentence; this has been observed in a number of processing studies (for ambiguous sentences see Frazier, 1987; Mecklinger et al., 1995; Schriefers et al., 1995; Bader and Meng, 1999, among others). Moreover, based on the Input Processing model, VanPatten (1996, 2004) assumes that many of the problems the L2 learners encounter during L2 acquisition are related to the way they parse the L2 input. According to VanPatten (1996, 2004), such an erroneous mechanism is the first-noun processing strategy, which predicts that the L2 learners have the tendency to analyse the first NP in a clause as its subject.<sup>12</sup> We believe that the L2 learners of our study employ the first-noun strategy to comprehend the L2 input but are not solely guided by this strategy, as suggested by their different performance on the SPM and the on-line tasks. Specifically, the overall accuracy of the L2 learners on the OSV sentences in the SPM tasks is 22%, whereas the corresponding percentage in the on-line task is 56%.<sup>13</sup> Notice also that the SPM task is demanding in terms of processing resources, as the participants listened to the sentences and they had to choose their response among four pictures. Based on these facts, we claim that the first-noun strategy affects the participants' performance especially when there is a processing load. That processing constraints affect the interpretation of these sentences is further confirmed by the NS data, which exhibit slower RTs for the sentences with non-canonical than with canonical word order in the on-line task. Another factor that might have resulted in the learners' low accuracy scores for the OSV sentences is that these sentences were not incorporated in an appropriate context which could have made their interpretation easier for the learners. To conclude, we argue that the scrambling sentences cause difficulties to our learners and these difficulties are related with processing factors. However, the data from the on-line task provide indication for the emergence of some sensitivity to the availability of OSV structures in Turkish.



In addition, the on-line data revealed very low accuracy rate for the  $O_{ABS}$  SV condition (38% across the three L2 groups), which shows that the L2 learners of our study accept scrambled sentences with unsuffixed objects and, thus, do not seem to be fully aware of the interaction between case morphology and word order constraints (contra Gürel, 2000; Haznedar, 2006). Furthermore, the fact that the judgements for the  $O_{ABS}$  SV sentences were the slowest in the L2 data, compared to the NS' rapid rejections of these sentences, provides evidence for the lack of automaticity in the access and the on-line use of the relevant features. A difference between our study and previous studies in L2 Turkish, which may account for the discrepancies between previous findings and our findings, is that we used a timed grammaticality judgement task, whereas the previous studies employed off-line tasks. It remains to be seen, therefore, whether the sensitivity to the interaction of Case and word order constraints found in the aforementioned studies would be replicated with on-line measures. Nevertheless, the RT data should not refrain us from concluding that case morphology and word order interact in the process of L2 acquisition. The fact that the  $O_{ABS}$  SV condition was significantly slower than the  $O_{ACC}$  SV condition shows that the learners treat the two conditions differently. This in turn indicates that there is some sensitivity to the interplay between Case and word order. Moreover, the trials in which the  $O_{ABS}$  SV sentences were judged as ungrammatical were significantly slower than those in which they were judged as grammatical (see Figure 3). This result signals the emergence of some awareness of the word order constraints although it does not provide evidence for their acquisition nor for automatic access to the relevant features. The lack of sensitivity to the interaction of Case and word order constraints is expected under the organic grammar, even though it is not clear to us whether the model would predict even the emergence of awareness of this interaction.

It is important to mention that these findings are not problematic for the Full Transfer/ Full Access theory, since the interplay of Case and word order is not attested in the first language. It is obvious, however, that the present data cannot be accounted for by the Morphological Deficit model because the problems observed do not seem to lie in the domain of surface morphology alone. Similarly, the Failed Functional Features and the Interpretability hypotheses fail to provide an explanation of the data because the relevant features are interpretable, i.e. referentiality, and grammaticalized in the L1, i.e. Case. Even though the three latter models attempt to capture the phenomena observed in the final stages of second language acquisition, we claim that the present data are problematic for these theories based on the observation that the learners perform much better on judging (un)grammaticalities related to verbal suffixes (see Table 8), which involve features that are also grammaticalized in the L1. In fact, the present data suggest that even interpretable features, such as referentiality, which are not grammaticalized in the L1, at least not in the same way as in the L2, will be problematic in the L2 development compared to features that are grammaticalized in the L1. More specifically, the accusative suffix in Turkish denotes specificity/referentiality, whereas the corresponding morpheme in Greek carries only Case features. This difference might be responsible for the difficulties the L2 learners exhibited. Needless to say, the present findings do not provide any insight on whether the L2 interpretable features, which are not grammaticalized in the L1, are acquirable.



## VIII Conclusions

The results of the present study on the acquisition of Turkish as a second language indicate considerable difficulties with Case morphology as well as problems with non-canonical word orders and with the interaction of Case and word order, even though the data suggest emergence or awareness of the relevant constraints. Our findings cannot be easily accommodated within a single model of the current L2 theories.

The high performance on verbal suffixes provides evidence for the existence of functional categories already at the early stages of L2 Turkish, contra the predictions of the organic grammar account. Moreover, the results indicate that L2 properties related to parameterization are acquired early, in accordance with the predictions of the Full Transfer/Full Access, the Morphological Deficit and the representational deficit accounts. Furthermore, we discovered that processing considerations seem to affect the learners' behaviour to a larger extent than they affect native speakers. There is no doubt that more research is needed in order to determine the exact conditions under which these considerations are in operation. For instance, it is still under debate whether they trigger the adoption of heuristics by the learners or not (cf. VanPatten, 1996, 2004; Clahsen and Felser, 2006) and how they interact with the development of the grammar.

Significantly, the data show that the development of inflectional morphology is problematic for L2 learners as suggested by the Morphological Deficit account although the morphological domains are not affected in the same way, i.e. there is a discrepancy between the nominal and the verbal domain. The interpretability of the features and their grammaticalization in the L1 cannot provide a complete explanation of the exact nature of the difficulties the participants of this study experienced. It may be the case that the features either that are not grammaticalized in the L1 or that are differently encoded in the morphosyntax of the L1 are more difficult to be acquired than those which are grammaticalized irrespective of their interpretability.

Finally, our results suggest that the specific properties of the morphological paradigms seem to affect the L2 development, even though more research is required in order to discover the factors that facilitate or inhibit the acquisition of L2 morphology.

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## Notes

- 1 Most prepositions in Greek take complements that are marked for accusative case with the exception of very few prepositions that come from the puristic form and take complements in genitive.
- 2 However, this is not always true, as there are cases in which generic objects are marked with the accusative suffix. Thus, a generic object may optionally carry the accusative suffix, when it is the object of certain verbs such as *sevmek* (= I like) and the subject is in plural. Moreover, the accusative marker is obligatory when the generic object is plural and when an indefinite object carries a possessive suffix (for detailed discussion on these issues and examples see Tura, 1973; Dede, 1986).
- 3 Notice that similar restrictions, which permit null formatives, are not observed in the verbal domain.
- 4 For a more detailed discussion on the factors that may affect word order in Turkish see Erguvanlı, 1984; Dede, 1986; Kornfilt, 1997; Keleş, 2001; Göksel and Kerslake, 2005.
- 5 Notice, however, that bare plurals in Greek are necessarily non-specific:
 

(i) o            janis            ayorase            pinakes.  
       the-NOM   John-NOM   buy-PAST.3SG   paintings-ACC  
       'John bought paintings.'
- 6 One reviewer raises the question whether specificity is indeed an interpretable feature. The accusative suffix in Turkish incorporates a specificity (Enç, 1991) or referentiality (Kılıçaslan, 2006) feature, which has implications on the way an accusative marked DP is linked to an antecedent in the discourse (Enç, 1991) but also on the pragmatic presuppositions it carries (for the pragmatic presuppositions of the accusative suffix in Turkish and their relation to the prosodic structure, see Nakipoğlu, 2009). Therefore, we consider the specificity feature as interpretable, since it has semantic effects on the interpretation of the DP and, consequently, on the sentential meaning.
- 7 One anonymous reviewer asked why there were only two gaps testing non-specific objects in the cloze task. We decided not to include more gaps for non-specific objects because non-specific objects in Turkish are marked by the null formative; thus, correct performance would coincide with omission and, hence, we would have difficulties in interpreting the results.
- 8 These sentences function as fillers in order to divert the subjects' attention from the experimental sentences. However, we call them pseudo-distractors, because they are expected to provide useful information regarding the subjects' performance on the items tested, which can be fruitfully analysed in the light of L2 theories of acquisition.
- 9 An anonymous reviewer points out that it is very unusual for native speakers to be 100% accurate in such a task. Notice, however, that the sentences in the sentence picture-matching task were very simple transitive sentences (e.g. 'a boy kisses a girl') and the speakers' task was to point to the right picture.
- 10 We would like to thank an anonymous reviewer who made this suggestion.
- 11 We thank an anonymous reviewer for raising this issue.
- 12 Notice that the SOV is also more frequent than the OSV order in Turkish (Erguvanlı, 1984) and this might have also affected the learners' performance. We thank an anonymous reviewer for raising this issue.

- 13 One anonymous reviewer suggests that the learners perform better in the on-line task, because the subject was animate and the object inanimate. Even though animacy is a cue that the learners may have employed, we do not think that this factor provides an entire explanation of their behaviour. If the learners had totally relied on animacy, they would have performed similarly in all four conditions of the on-line task. The accuracy as well as the RT data show that this was not the case.

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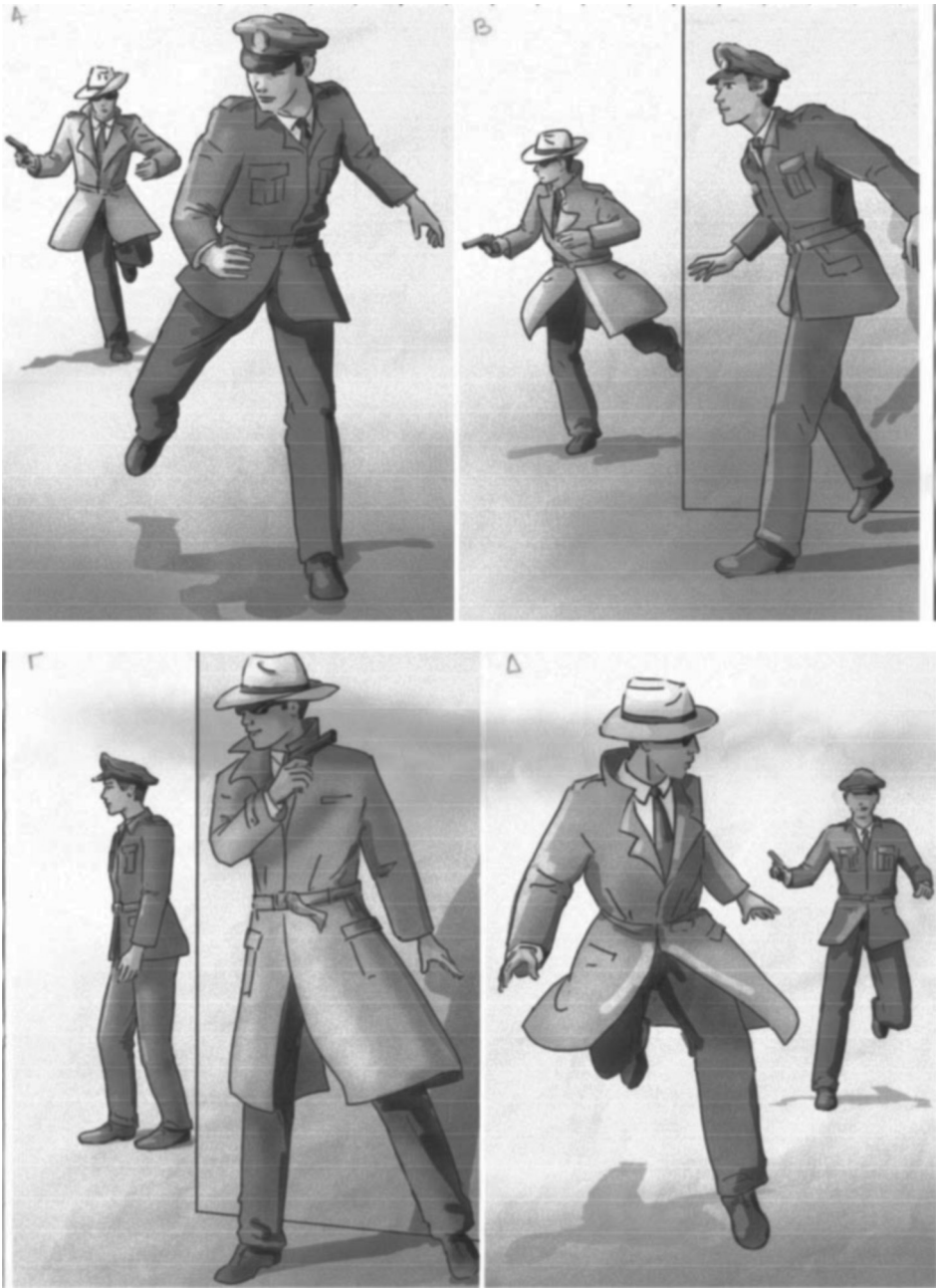
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**Appendix I** A quartet of pictures used in the SPM task for the item *Gangster polisi kovalıyor* (gangster policeman-ACC is-chasing)





**Appendix 2** The entire list of the experimental items used in the on-line task is provided below. An example of the way the four conditions were constructed is shown in (1a)–(1d)

- |        |                  |                  |                 |
|--------|------------------|------------------|-----------------|
| 1. (a) | Çocuk            | gazete-ler-i     | oku-du.         |
|        | boy              | newspaper-PL-ACC | read-PAST       |
| (b)    | Gazete-ler-i     | çocuk            | oku-du.         |
|        | newspaper-PL-ACC | boy              | read-PAST       |
| (c)    | Çocuk            | gazete           | oku-du.         |
|        | boy              | newspaper        | read-PAST       |
| (d)    | *Gazete          | çocuk            | oku-du.         |
|        | newspaper        | boy              | read-PAST       |
| 2.     | Kız              | mektub-u         | yaz-dı.         |
|        | girl             | letter-ACC       | write-PAST      |
| 3.     | Adam             | çanta-yı         | at-tı.          |
|        | man              | bag-ACC          | throw away-PAST |
| 4.     | Öğrenci          | bilgisayar-ı     | al-dı.          |
|        | student          | computer-ACC     | buy-PAST        |
| 5.     | Hasta            | hastane-yi       | gör-dü.         |
|        | patient          | hospital-ACC     | see-PAST        |
| 6.     | Çocuk            | yemeğ-i          | ye-di.          |
|        | boy              | food-ACC         | eat-PAST        |
| 7.     | Manav            | meyve-ler-i      | sat-tı.         |
|        | grocer           | fruit-PL-ACC     | sell-past       |
| 8.     | Baba             | televizyon-u     | kapa-dı.        |
|        | father           | television-ACC   | turn off-PAST   |
| 9.     | Gazeteci         | araba-yı         | al-dı.          |
|        | journalist       | car- ACC         | take-PAST       |
| 10.    | Kadın            | valiz-i          | hazırla-dı.     |
|        | woman            | suitcase-ACC     | prepare-PAST    |
| 11.    | Anne             | kahvaltı-yı      | hazırla-dı.     |
|        | mother           | breakfast-ACC    | prepare-PAST    |
| 12.    | Öğretmen         | kitab-ı          | bul-du.         |
|        | teacher          | book-ACC         | find-PAST       |
| 13.    | Çocuk            | defter-i         | aç-tı.          |
|        | boy              | notebook-ACC     | open-PAST       |
| 14.    | Kız              | kolye-yi         | ver-di.         |
|        | girl             | necklace-ACC     | give-PAST       |
| 15.    | Büyükanne        | ceket-i          | temizle-di.     |
|        | granmama         | cardinal-ACC     | clean-PAST      |
| 16.    | Çocuk            | pirzola-yı       | ye-di.          |
|        | boy              | steak-ACC        | eat-PAST        |
| 17.    | Müşteri          | salata-yı        | iste-di.        |
|        | customer         | salad-ACC        | ask for-PAST    |
| 18.    | Turist           | taverna-yı       | ar-ıyor.        |
|        | tourist          | tavern-ACC       | look for-PRES   |

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|-----|---------|------------|------------|
| 19. | Kedi    | top-u      | bul-du.    |
|     | cat     | ball-ACC   | find-PAST  |
| 20. | Adam    | bira-yı    | iç-ti.     |
|     | man     | beer-ACC   | drink-PAST |
| 21. | Çocuk   | otobüs-ü   | gör-dü.    |
|     | boy     | bus-ACC    | see-PAST   |
| 22. | Öğrenci | pencere-yi | aç-tı.     |
|     | student | window-ACC | open-PAST  |
| 23. | Baba    | peçete-yi  | ver-di.    |
|     | father  | towel-ACC  | give-past  |
| 24. | Bakkal  | peynir-i   | ye-di.     |
|     | grocer  | cheese-ACC | eat-PAST   |